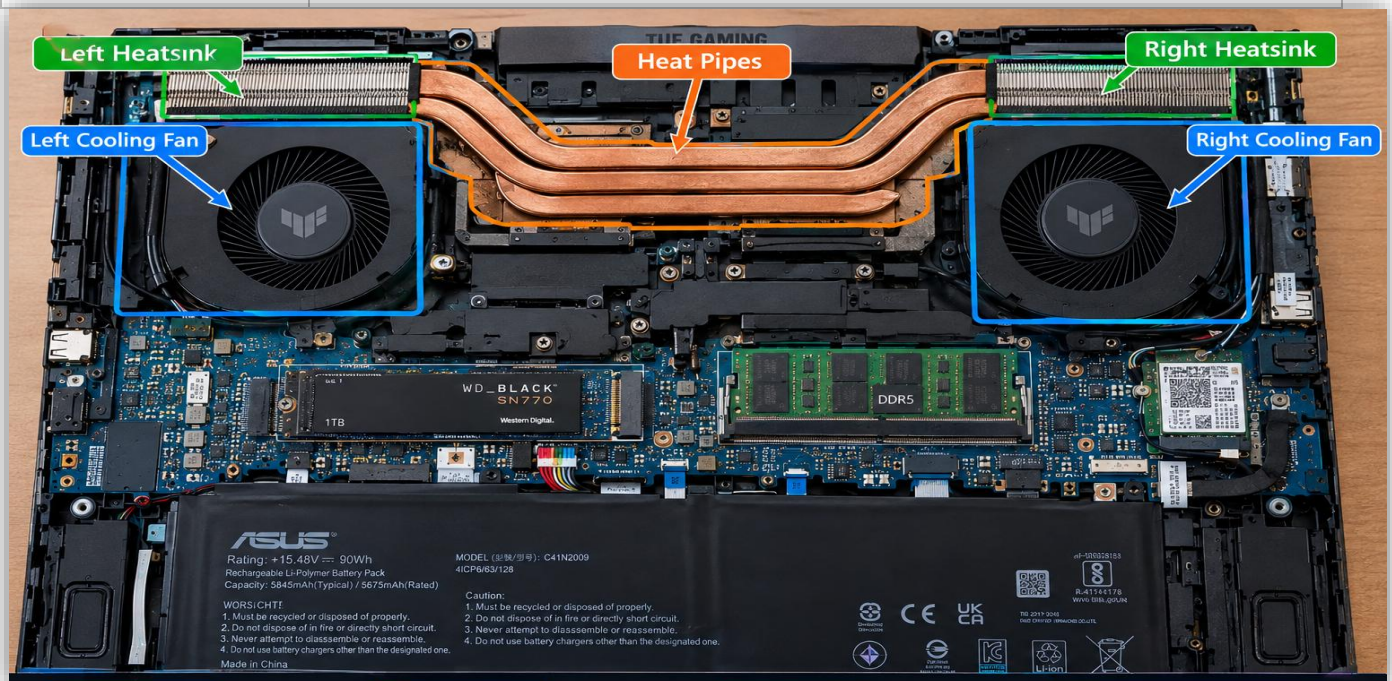


1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

<i>Item</i>	<i>Details</i>
<i>GPU Brand</i>	NVIDIA
<i>Model Number</i>	GeForce RTX 3060
<i>Type</i>	Dedicated GPU
<i>Laptop</i>	ASUS TUF Gaming F17

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

Component	Function
Cooling Fans	Draw cool air into the laptop and push hot air out.
Heat Pipes	Transfer heat from the CPU/GPU to the heatsinks.
Heatsinks	Dissipate heat into the surrounding air through metal fins.
Thermal Paste	Improves heat transfer between the CPU/GPU and heatsink.



3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Factor	Air Cooling	Liquid Cooling (AIO / Custom Loop)
Initial Cost	Low to Moderate: Factory-installed heatsinks and fans add minimal cost to the graphics card.	High to Extreme: AIO GPU solutions add a steep premium, while custom loops require separate water blocks, radiators, and pumps.
Thermal Efficiency & Noise	Good: Effective for standard TDP loads, but fans must spin at higher RPMs under full load, producing audible noise.	Superior: Water has much higher specific heat capacity than air, keeping boost clocks higher while radiator fans spin at lower, quieter speeds.
Maintenance & Reliability	Low: Virtually maintenance-free; only requires occasional compressed air cleaning to clear dust buildup.	Moderate to High: AIOs face pump wear or permeation over 3–6 years; custom loops require regular fluid flushes, leak tests, and biocide top-ups.
Failure Risk	Low Risk: If a fan motor fails, it can easily be swapped out with zero threat of fluid spills.	Catastrophic Risk Potential: Pump failure causes immediate thermal throttling; fitting/tube leaks risk frying expensive electronics.

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

- I would choose **liquid cooling** for the GPU.
- It provides efficient heat dissipation during long gaming and streaming sessions, helping maintain stable performance.
- Although liquid cooling is more expensive and requires more maintenance than air cooling, it is suitable for a high-performance gaming PC.
- For a lower-budget build, however, a good air-cooling system would also be sufficient.